Trainings

General Information

Industrial Applications

For your convenience, listed below are the section numbers that contain specific instructions for performing the maintenance checks listed in the maintenance schedule.

Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous checks that are due for scheduled maintenance.

Maintenance Procedures at Daily Interval[4] (Section 3)

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Fan, Cooling - Check
- Coolant Level - Check
- Air Intake Piping - Check[4]
- Air Tanks and Reservoirs - Drain[4]
- Drive Belts - Check

Maintenance Procedures at 250 Hours, or 3 Months[1, 2, 4] (Section 5)

- Fuel Filter (Spin-On Type) - Change
- Lubricating Oil and Filters - Change[1]
- Charge-Air Cooler - Check[4]
- Charge-Air Piping - Check[4]
- Air Cleaner Restriction - Check[4]
- Air Compressor - Check[4]
- Radiator Pressure Cap - Check

Maintenance Procedures at 500 Hours, or 6 Months[2, 3, 4] (Section 7)

- Coolant Filter, if equipped - Change
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[2, 3]
- Air Compressor Discharge Lines - Check[4]

Maintenance Procedures at 1000 Hours, or 1 Year[4] (Section 9)

- Overhead Set - Adjust
- Cooling Fan Belt Tensioner - Check
- Air Cleaner Assembly (Engine-Mounted) - Change[4]

Maintenance Procedures at 2000 Hours, or 2 Years[2, 3, 4, 5] (Section 10)

- Vibration Damper, Rubber - Inspect for Reuse

- Vibration Damper, Viscous - Inspect for Reuse
- Cooling System - Flush[2, 3, 5]
- Air Compressor Discharge Lines - Check[4]

1. The lubricating oil and lubricating oil filter interval can be adjusted based on fuel consumption, gross vehicle weight, and idle time. Refer to Oil Drain Intervals in this section.
2. Test the SCA concentration level every 6 months unless concentration is over three units; then check at every oil drain interval until concentration is below three units.
3. Antifreeze check interval is every oil change or 500 hours or 6 months, whichever occurs first. The operator **must** use a heavy-duty year-round antifreeze that meets the chemical composition of ASTM D6210. The antifreeze change interval is 2 years. Antifreeze is essential for freeze, overheat, and corrosion protection.
4. Follow the manufacturer's recommended maintenance procedures for the starter, alternator, generator, batteries, electrical components, engine brakes, exhaust brake, charge-air cooler, air compressor, refrigerant compressor, and fan clutch.
5. This cooling system requirement to Flush at this scheduled maintenance includes Drain, Flush, and Fill.

Generator Set Applications

For your convenience, listed below are the section numbers that contain specific instructions for performing the maintenance checks listed in the maintenance schedule.

Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous checks that are due for scheduled maintenance.

Maintenance Procedures at Daily Interval[1] (Section 3)

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Fan, Cooling - Check
- Coolant Level - Check
- Air Intake Piping - Check[4]
- Air Tanks and Reservoirs - Drain[4]
- Drive Belts - Check

Maintenance Procedures at 300 Hours, or 6 Months[1, 2] (Section 4)

- Fuel Filter (Spin-On Type) - Change
- Charge-Air Cooler - Check
- Air Cleaner Restriction - Check
- Fuel Supply Lines – Check
- Fuel Drain Lines – Check
- Coolant Filter – Change [1,2]

Maintenance Procedures at 1500 Hours, or 1 Year [3] (Section 5)

- Drive Belts - Check
- Overhead Set - Adjust

Maintenance Procedures at 6000 Hours, or 2 Years [1, 2, 3] (Section 6)

- Cooling System - Drain, Flush, and Fill [1, 2]
- Vibration Damper, Rubber - Check
- Vibration Damper, Viscous - Check

1. Test the SCA concentration level every 6 months if concentration is below three units; check at every oil drain interval if the concentration is above three units.
2. Antifreeze check interval is every oil change or 500 hours or 6 months, whichever occurs first. The operator **must** use a heavy-duty year-round antifreeze that meets the chemical composition of ASTM D6210. The antifreeze change interval is 2 years. Antifreeze is essential for freeze, overheat, and corrosion protection.
3. Follow the manufacturer's recommended maintenance procedures for the starter, alternator, generator, batteries, electrical components, engine brakes, exhaust brake, charge-air cooler, air compressor, refrigerant compressor, and fan clutch.

Marine Applications

Maintenance Procedures at Daily Interval[1] (Section 3)

- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Marine Gear - Check[1]
- Drive Belts - Check

Maintenance Procedures at 75 Hours or 3 Months[3] (Section 4)

- Zinc Anode - Check[3]
- Cooling System Hoses - Check
- Sea Water Hoses - Check
- Air Cleaner Restriction - Check
- Batteries - Check
- Battery Cables and Connections - Check
- Component Connector and Pin Inspection - Check

Maintenance Procedures at 300 Hours or 1 Year[1, 2] (Section 6)

- Fuel Filter (Spin-On Type) - Change
- Fuel-Water Separator Element - Replace
- Lubricating Oil and Filters - Change[2]
- Coolant Filter, if equipped - Change
- Engine Coolant Heater - Check
- Marine Gear Oil Cooler - Flush[1]
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[2]
- Heat Exchanger - Flush
- Sea Water Pump - Replace
- Aftercooler Assembly (Sea Water) - Flush

- Air Cleaner Assembly (Engine-Mounted) - Check
- Air Intake Piping - Check
- Marine Gear Oil - Check[1]
- Radiator Pressure Cap - Check
- Engine Wiring Harness - Check

Maintenance Procedures at 600 Hours or 2 Years (Section 8)

- Vibration Damper, Rubber - Inspect for Reuse
- Vibration Damper, Viscous - Inspect for Reuse
- Overhead Set - Adjust
- Cooling System - Flush[4]
- Cooling Fan Belt Tensioner- Check

1. Consult the marine gear manufacturer operator's manual for specifications and recommendations.
2. Refer to Refer to Procedure 018-024 in Section V. (/qs3/pubsys2/xml/en/procedures/41/41-018-024.html)
3. Depending upon the quality of electrical bonding and water conditions, increased maintenance is sometimes necessary.
4. This cooling system requirement to Flush at this scheduled maintenance includes Drain, Flush, and Fill.

Oil Drain Intervals

Refer to the following flowchart to determine the maximum recommended oil change and filter change intervals in kilometers, miles, hours, or months, whichever comes first.

Is the vehicle one of those listed below?

- Truck crane/yard spotter
- Paver/crane/backhoe
- Dozer/scrape/skipper

If Yes -

- Select the correct oil drain interval from Table 1.

If No -

- Is the vehicle one of those listed below?
  - Tractor/combine/irrigation equipment
  - Generator set/air compressor/fire equipment

If Yes -

- Select the correct oil drain interval from Table 2.

If No -

- Select the correct oil drain interval from Table 3.

Table 1, Oil Drain Intervals*				
Vehicle/Equipment	**Kilometers**	**Miles**	**Hours**	**Months**
Truck crane/yard spotter	10,000	6,000	250	3
Paver/crane/backhoe	N/A	N/A	250	3
Dozer/scraper/skidder	N/A	N/A	250	3

Table 2, Oil Drain Intervals*				
Vehicle/Equipment	**Kilometers**	**Miles**	**Hours**	**Months**
Tractor/combine/irrigation equipment	N/A	N/A	250	3
Generator set/air compressor/fire pump	N/A	N/A	250	3

Table 3, Oil Drain Intervals*				
Vehicle/Equipment	**Kilometers**	**Miles**	**Hours**	**Months**
All others	10,000	6,000	250	3

*Units equipped with a 26.5 liter [28 qt] high capacity oil pan can extend intervals to 500 hours.